Lab Safety

- Why should malware never be executed on a personal computer? Because it can infect the system, steal data, damage files, or spread to other devices.

- What is the purpose of a malware analysis sandbox? To analyze malware safely in an isolated environment.

- Why are snapshots useful during malware analysis? They allow analysts to restore the virtual machine to its previous state.

- Why is network isolation important? It prevents malware from spreading or communicating with external systems.

Malware Classification

- What is the difference between a virus and a worm? A virus attaches itself to a file or program, while a worm spreads independently across systems or networks.

- How does a Trojan rely on user deception? It disguises itself as legitimate software to trick users into installing or executing it.

- What is ransomware designed to do? To encrypt files or lock systems and demand a ransom.

- Why are rootkits difficult to detect? Because they hide their presence and malicious activities within a system.

- What is a botnet? A network of compromised devices controlled by an attacker.

Malware Samples

- Why should analysts use trusted malware repositories? To obtain malware samples and related information from reliable sources.

- What information can be found in malware sample metadata? File name, size, type, timestamps, hashes, and other file properties.

- Why are file hashes important? They help identify files and compare them against known malware samples.

- What is a malware family? A group of malware variants that share similar code, behavior, or characteristics.

Analysis Techniques

- What is the difference between static and dynamic analysis? Static analysis examines malware without executing it, while dynamic analysis observes its behavior during execution.

- Why is static analysis performed before execution? To gather information about the malware and identify potential risks before running it.

- What information can strings reveal? URLs, IP addresses, file paths, commands, error messages, and suspicious text.

- What types of behavior are monitored during dynamic analysis? File changes, process creation, registry modifications, network connections, and system activity.

- Why do analysts use sandbox environments? To observe malware behavior in a controlled and isolated environment.

Indicators of Compromise (IOCs)

- What is an IOC? An Indicator of Compromise is evidence that a system may have been compromised.

- Give examples of common IOCs. File hashes, malicious IP addresses, suspicious domains, URLs, file paths, and unusual network connections.

- Why are hashes useful during investigations? They help identify suspicious files and compare them with known malware samples.

- Why should organizations collect multiple indicators instead of relying on one? Because one indicator may be insufficient or change over time, while multiple indicators provide stronger evidence.

- How do IOCs help security teams? They help detect threats, investigate incidents, identify compromised systems, and block malicious activity.